Andrew Carnes

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Education

University of Central Florida Expected Graduation: Spring 2028
B.S. Mechanical Engineering, Burnett Honors College *GPA: 3.69*

Experience

Lockheed Martin Rotary and Mission Systems June 2026 – Present
Quality Engineering CWEP (College Work Experience Program)

- Built a shared **Excel** database mapping defective-material flow across the subassemblies area, eliminating time spent searching bins and racks for parts and enabling the closure of 51 quality notifications (Q-Notes) in a single day.
- Lead daily quality assurance stand-ups for 10+ engineers, inspectors, and operators, presenting defect trends and projections so every new Q-Note is addressed on creation, cutting open Q-Notes by 38% in 6 weeks.
- Drive **root cause and corrective action (RCCA)** by writing corrective action reports for supervisors and colleagues, supporting conformance to **AS9100** and **IPC/WHMA-A-620** standards.

Consolidated Electrical Distributors January 2026 – May 2026
Project Management Intern

- Programmed and created documentation for an **Excel VBA** tool that generates standardized customer shipping-status reports from mapped vendor data, eliminating transfer errors and cutting entry time by 99.5%.
- Led branch-wide deployment of a new warehouse management system by developing an alphanumeric bin location system and standardizing product and location data to enable scan-based inventory management of 6,480 SKUs.
- Administered the VBA tool to support logistics tracking and reporting for a $2.32M commercial healthcare construction project, ensuring accurate shipment and delivery data.

Gas-Regolith Interaction Testbed (GRIT), University of Central Florida August 2025 – Present
Undergraduate Researcher, CFD Simulations Lead

- Model over-, perfectly, and under-expanded plumes from 3 team-designed nozzles in **ANSYS Fluent** to quantify how ambient pressure affects plume collimation, completing 6 steady-state cases across lunar and Martian conditions.
- Process crater-profile data for dynamic-mesh Fluent simulations of the evolving regolith surface and run transient plume simulations, with the first Mars case complete.
- Lead weekly technical discussions for 5 newly onboarded researchers, training them on the simulation workflow.
- Research plume-induced regolith erosion and high-velocity ejecta that threaten nearby hardware at co-located landing sites.

Projects

Knights Experimental Rocketry, University of Central Florida August 2025 – June 2026
Aerostructures Lead (later Team Member)

- Led a 12-member team in **composite layup** and manufacturing of the "Odysseus" rocket airframe for the 2026 International Rocket Engineering Competition.
- Wrote the team's Manufacturing Process Plans for **carbon-fiber and fiberglass wet layup** of the nosecone, body tubes, and structural components, cutting layup time by 36%.
- Coordinated cross-team hardware integration between propulsion, recovery, controls, and payload systems; reached a 9,935 ft apogee, within 0.65% of the 10,000 ft target.

NASA L'SPACE Proposal Writing and Evaluation Experience Academy January 2025 – April 2025
Project Manager

- Directed a 10-member multidisciplinary national team and co-authored PEDALS, an official NASA Marshall Space Flight Center **technical proposal** for a magnetorheological (MR) damping off-planet landing system.
- Improved theoretical landing-velocity tolerance 2.5x over state-of-the-art systems (2.4 to 6.1 m/s) by pairing terrain sensing with MR dampers and adaptive landing legs.
- Led the review of 1 of 3 peer proposals on a 12-member panel using NASA-defined criteria (technical approach, feasibility, resource alignment), delivering structured feedback and scoring recommendations.

Skills

CAD/Engineering: SolidWorks, Siemens NX, Autodesk Inventor, Fusion 360, ANSYS Fluent, OpenRocket
Programming/Data: Python, MATLAB, Arduino, Microsoft Excel, Excel VBA, SAP, Tableau
Manufacturing: 3D Printing, Composite Layup (Carbon Fiber, Fiberglass), Mechanical Assembly & Drawings, Prototyping, Jig Design, Introductory CNC Machining, Root Cause Corrective Action, AS9100, IPC/WHMA-A-620
Certifications: CSWA, CSWP, Autodesk Inventor Certified, Microsoft Excel Certified, Siemens NX (In Progress)
Interests: State Champion Rower | Jazz & Classical Pianist | Project Downtown Orlando Volunteer